Summary in Graph

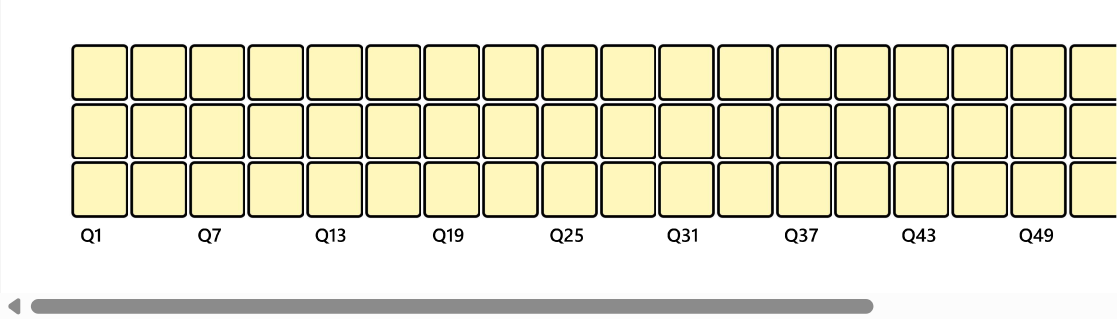
# Exam Summary (GO Classes DA Test Series 2026 | MOCK GATE | Test 3)

| | | | |
|---|---|---|---|
| Qs. Attempted: | 0 (0 + 0) | Correct Marks: | 0 (0 + 0) |
| Correct Attempts: | 0 (0 + 0) | Penalty Marks: | 0 (0 + 0) |
| Incorrect Attempts: | 0 (0 + 0) | Resultant Marks: | 0 (0 + 0) |

| | |
|---|---|
| Total Questions: | 65 (30 + 35) |
| Total Marks: | 100 (30 + 70) |
| Exam Duration: | 180 Minutes |
| Time Taken: | 0 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (25) out of 251 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

## Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Charlie is making a necklace with yellow beads and green beads. She has already used 4 green beads and 0 yellow beads. How many yellow beads will she have to add so that $\frac{4}{5}$ of the total number of beads are yellow?

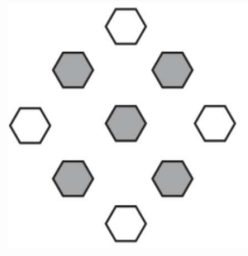
A. 16
B. 4

C. 20
D. 24

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 03sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Spatial Aptitude

In the diagram, some of the hexagons are shaded.



What fraction of all of the hexagons are shaded?

A. $\frac{1}{2}$ B. $\frac{5}{9}$ C. $\frac{4}{5}$ D. $\frac{1}{3}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

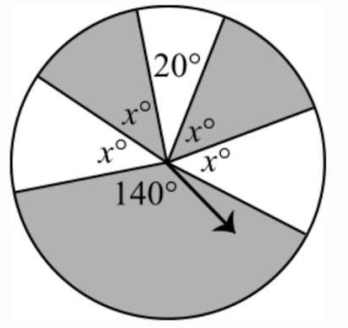
Kamal turned his computer on at 2 p.m. on Friday. He left his computer on for exactly 30 consecutive hours. At what time did he turn his computer off?

A. 4 p.m. on Saturday
B. 6 p.m. on Saturday
C. 8 p.m. on Sunday
D. 8 p.m. on Saturday

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

A circular spinner is divided into six regions, as shown. Four regions each have a central angle of $x^\circ$. The remaining regions have central angles of $20^\circ$ and $140^\circ$. An arrow is attached to the centre of the circle.




The arrow is spun once. What is the probability that the arrow stops on a shaded region?

A. $\frac{2}{3}$ B. $\frac{7}{8}$ C. $\frac{1}{2}$ D. $\frac{5}{12}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair:

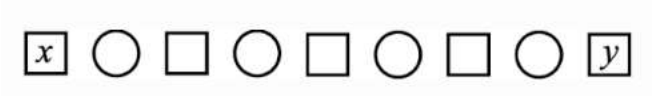
**Gladiator : Arena**

A. dancer : stage
B. commuter :train
C. teacher :classroom
D. lawyer :courtroom

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

The integers $1, 2, 4, 5, 6, 9, 10, 11, 13$ are to be placed in the circles and squares below with one number in each shape.




Each integer must be used exactly once and the integer in each circle must be equal to the sum of the integers in the two neighbouring squares. If the integer $x$ is placed in the leftmost square and the integer $y$ is placed in the rightmost square, what is the largest possible value of $x + y$?

A. 24
B. 20
C. 22
D. 21

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Jiwei and Hari entered a race. Hari finished the race in $\frac{4}{5}$ of the time it took Jiwei to finish. The next time that they raced the same distance, Jiwei increased his average speed from the first race by $x\%$, while Hari maintained the same average speed as in the first race. In this second race, Hari finished the race in the same amount of time that it took Jiwei to finish. The value of $x$ is _____.

A. 20
B. 25
C. 35
D. 40

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Zebadiah has 3 red shirts, 3 blue shirts, and 3 green shirts in a drawer. Without looking, he randomly pulls shirts from his drawer one at a time. He would like a set of shirts that includes either 3 of the same colour or 3 of different colours. What is the minimum number of shirts that Zebadiah has to pull out to guarantee that he has such a set?

A. 4
B. 3
C. 6
D. 5

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

Hagrid has 100 animals. Among these animals,

- each is either striped or spotted but not both,
- each has either wings or horns but not both,
- there are 28 striped animals with wings,
- there are 62 spotted animals, and
- there are 36 animals with horns.

How many of Hagrid's spotted animals have horns?

A. 8
B. 10
C. 2
D. 26

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

Creative minds see new angles to everyday things that most people would take for granted; indeed there is always a sense of _____ or wonder at even the most mundane objects.

A. pride
B. anxiety
C. surprise
D. curiosity

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

# Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Suppose we use merge sort to sort the array $[9, 1, 1, 3, 5, 5, 6, 8]$. During the final step of merge sort, two sorted arrays are merged. What are the two arrays that will be merged in the final step?

A. $[1, 1, 3, 5]$ and $[5, 6, 8, 9]$
B. $[1, 1, 3, 9]$ and $[5, 5, 6, 8]$
C. $[1, 3, 5, 5]$ and $[1, 6, 8, 9]$
D. $[1, 1, 5, 9]$ and $[3, 5, 6, 8]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

If $A$ and $B$ are mutually exclusive events with $P(A) = 0.6$ and $P(B) = 0.3$, what will be the value of $P(B' \mid A')$?

A. 0.10
B. 0.40
C. 0.25
D. 0.50

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Suppose $X$ is an exponential random variable with rate $\lambda = 1$, where $P(X > a) = e^{-\lambda a}$. Let $Y = e^{-X}$. What is the cumulative distribution function (CDF) of $Y$ for $t \in [0, 1]$?

A. $P(Y \leq t) = 1 - t$
B. $P(Y \leq t) = t^2$
C. $P(Y \leq t) = t$
D. $P(Y \leq t) = -\ln(t)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Which of the following expressions is equal to $\mathrm{P(A, B, C, D)}$ without assuming independence?

A. $P(D|A, B, C).\, P(A, B, C)$
B. $P(D|A, B, C).\, P(A).\, P(B) \times P(C)$
C. $P(A|B).\, P(B|C).\, P(C|D).\, P(D)$
D. $P(A|B, C, D).\, P(B|C, D).\, P(C|D).\, P(D)$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Suppose $X$ and $Y$ are independent standard normal random variables $(N(0, 1))$. Recall that for independent random variables, the expectation of their product is the product of their expectations.

What is the value of $E\left[X^2Y^2\right]$?

A. 0
B. 1
C. 2
D. 0.5

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Probability

Consider two continuous random variables $X$ and $Y$ that are uniformly distributed with their joint PDF equal to $A$ over the shaded region shown in the figure below. What is the value of $A$?

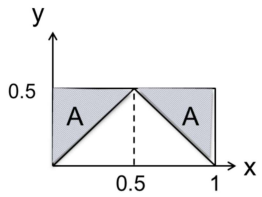



Figure 1: Joint pdf of $X$ and $Y$

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $T : R^4 \rightarrow R^3$ be a linear transformation such that:

$$T\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 - x_4 \\ 3x_2 - x_3 + 2x_4 \\ x_1 + x_4 \end{bmatrix}.$$

Find the matrix $A$ of $T$.

A. $A = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 3 & -1 & 2 \\ 1 & 0 & 0 & 1 \end{bmatrix}$

B. $A = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 3 & 1 & 2 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

C. $A = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 3 & -1 & 2 \\ 1 & 0 & 1 & -1 \end{bmatrix}$

D. $A = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 3 & -1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Which of the following represents the correct basis for the null space $N(\boldsymbol{A})$ of the matrix

$$\boldsymbol{A} = \begin{bmatrix} 1 & -3 & 0 \\ 2 & -6 & 4 \\ -3 & 9 & 1 \end{bmatrix}?$$

A. $N(\boldsymbol{A}) = \text{span}\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right\}$

B. $N(\boldsymbol{A}) = \text{span}\left\{\begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}\right\}$

C. $N(\boldsymbol{A}) = \text{span}\left\{\begin{bmatrix} 14 \\ -1 \\ 4 \end{bmatrix}\right\}$

D. $N(\boldsymbol{A}) = \text{span}\left\{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let

$$A = \begin{pmatrix} 2 & t & 0 \\ t & 2 & t \\ 0 & t & 2 \end{pmatrix}$$

Find all values of $t$ for which the matrix $A$ has 3 negative eigenvalues.

A. $t > 2$
B. $t < -2$
C. $t = 0$
D. No value of $t$ allows $A$ to have 3 negative eigenvalues

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

If $x, y, z \in \mathbb{R}^n$ are $n$-component vectors, the number of arithmetic operations required to compute $xy^T z$ scales as which of the following, depending on the order of computation?

A. $\left(xy^T\right) z \sim n, x\left(y^T z\right) \sim n^2$
B. $\left(xy^T\right) z \sim n^2, x\left(y^T z\right) \sim n^2$
C. $\left(xy^T\right) z \sim n^2, x\left(y^T z\right) \sim n$
D. $\left(xy^T\right) z \sim n^3, x\left(y^T z\right) \sim n^3$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

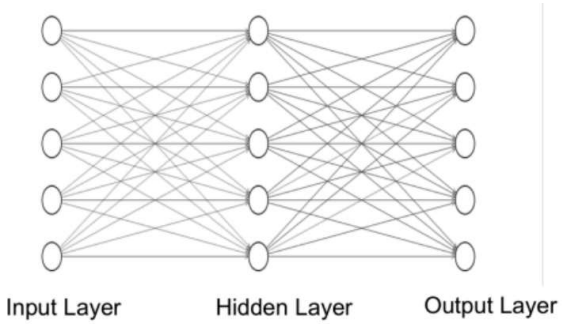
Let $A$ be a symmetric $3 \times 3$ matrix such that $\det(A) = -6$ and $\text{tr}(A) = 4$. Determine its definiteness.

A. A is positive definite.
B. $A$ is negative definite.
C. $A$ is indefinite.
D. $A$ is positive semidefinite.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

What are the number of parameters in a 2-layer neural network with 5 neurons in each layer as shown in figures?




Your Answer: | Correct Answer: 60 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Suppose we have a regularized linear regression model:

$$\text{argmin}_w \|Y - Xw\|_2^2 + \lambda \|w\|_1.$$

What is the effect of increasing $\lambda$ on bias and variance?

A. Increases bias, increases variance
B. Increases bias, decreases variance
C. Decreases bias, increases variance
D. Decreases bias, decreases variance

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Which of the following statements are true for Principal Components analysis

A. The solution to principal component analysis (PCA) can always be found using singular value decomposition
B. The features selected by PCA are linear combinations of the original features
C. PCA can be formulated as an optimization problem that finds the (orthogonal) directions of maximum covariance of a set of observations $X$
D. Principal component analysis is used to expand the dimension of features and increase the total variance within the data

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Convexity is a desirable property in machine learning because it:

A. guarantees gradient descent finds a global minimum in optimization problems for functions that have a global minimum
B. helps to avoid the model overfitting
C. speeds up model training
D. reduces model complexity

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Consider the properties and behavior of gradient descent in optimization problems. Which of the following statements are valid?

A. Initial parameter values have no impact on the convergence of gradient descent.
B. Convergence of gradient descent to a local minimum guarantees convergence to the global minimum.
C. The learning rate significantly influences the convergence rate of gradient descent.
D. Gradient descent can fail to converge if the learning rate is excessively high.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What are the contents of the list lst after the following code is executed?

```
lst = [2 * x + y for x in range(3) for y in range(3, 6)]
```

A. $[3, 4, 5, 6, 7, 8, 9]$
B. $[3, 4, 5, 5, 6, 7, 7, 8, 9]$
C. $[3, 5, 7, 4, 6, 8, 5, 7, 9]$
D. $[6, 7, 8, 8, 9, 10, 10, 11, 12]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Suppose $R$ and $S$ are two relations. Let $C$ be the referential integrity constraint that says: whenever $R$ has a tuple with some values $v_1, v_2, \ldots, v_n$ in particular attributes $A_1, A_2, \ldots, A_n$, there must be a tuple of $S$ that has the same values $v_1, v_2, \ldots, v_n$ in particular attributes $B_1, B_2, \ldots, B_n$.

Which of the following operations on $R, S$ can cause violation of the constraint $\mathrm{C}$?

A. Insert into $\mathrm{R}$
B. Insert into $\mathrm{S}$
C. Delete from $\mathrm{R}$
D. Delete from $\mathrm{S}$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Which of the following statements is/are FALSE?

A. Depth first search will find an optimal path with respect to the cost of the path.
B. Depth first search will find an optimal path with respect to the number of steps in the path.
C. Breadth first search will find an optimal path with respect to the cost of the path.
D. Breadth first search will find an optimal path with respect to the number of steps in the path.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Consider some convex function $f : \mathbb{R}^d \to \mathbb{R}$, and assume that $f$ is twice-differentiable. What can we say about $\nabla_x^2 f(x) \in \mathbb{R}^{d \times d}$, the Hessian of $f(x)$ ? (Hint: think about the case when $d = 1$, and what the second derivative of $f$ says about its shape.)

A. $\nabla_x^2 f(x)$ is negative semi-definite.
B. $\nabla_x^2 f(x)$ is negative definite.
C. $\nabla_x^2 f(x)$ is positive definite.
D. $\nabla_x^2 f(x)$ is positive semi-definite.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Consider some function $f(x) : \mathbb{R}^d \to \mathbb{R}$, and assume that we want to run an iterative algorithm to find the maximizer of $f$. Which update rule should we use to do this (for some $\eta > 0$)?

A. $x_{t+1} \leftarrow -x_t + \eta \cdot \nabla_x f(x_t)$.
B. $x_{t+1} \leftarrow x_t - \eta \cdot \nabla_x f(x_t)$.
C. $x_{t+1} \leftarrow -x_t - \eta \cdot \nabla_x f(x_t)$.
D. $x_{t+1} \leftarrow x_t + \eta \cdot \nabla_x f(x_t)$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 1 | Penalty: 0 | Calculus

Which of the following are true about a convex function $f(x) : \mathbb{R}^d \to \mathbb{R}$?

A. $f(x)$ must be differentiable across its entire domain.
B. $f(x)$ has a unique global minimum.
C. $g(x) = -f(x)$ is also convex.
D. If $f(x)$ is twice differentiable, then $z^\top \nabla^2 f(x) z \geq 0$ for all $z \in \mathbb{R}^d$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Which of the following statements is the negation of $\forall \boldsymbol{x} \exists \boldsymbol{y} \forall \boldsymbol{z}((\boldsymbol{P}(\boldsymbol{x}) \wedge \boldsymbol{Q}(\boldsymbol{y})) \rightarrow \boldsymbol{R}(\mathbf{z}))$?

A. $\forall x \exists y \forall z \sim ((P(x) \wedge Q(y)) \rightarrow R(z))$.
B. $\exists x \forall y \exists z(\sim (P(x) \wedge Q(y)) \vee R(z))$.
C. $\exists x \forall y \exists z(\sim R(z) \rightarrow \sim (P(x) \wedge Q(y)))$.
D. $\exists x \forall y \exists z(P(x) \wedge Q(y) \wedge \sim R(z))$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Which of the following statements is/are true:

A. if a propositional formula is valid then it is satisfiable.
B. if a propositional formula is not valid then it is satisfiable.
C. if a propositional formula is not valid then its negation is satisfiable.
D. if a propositional formula is not valid, then its negation is valid.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

A linear-probing hash table of length $10$ uses the hash function $\mathrm{h}(\mathrm{x}) = \mathrm{x} \mod 10$. After inserting six integer keys into an initially empty hash table, the array of keys appears as follows:

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| | 11 | 22 | 12 | 13 | 14 | 16 | | | |

Which of the following insertion sequences could result in the above hash table?

A. $11, 22, 12, 13, 14, 16$
B. $22, 11, 12, 13, 14, 16$
C. $12, 11, 22, 13, 14, 16$
D. $11, 12, 13, 14, 16, 22$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider an initial array of unsorted symbols:$[@, \#, \$, \%, \&, *]$, where no two symbols are considered equal. During the execution of insertion sort in increasing order, the array reaches the intermediate state: $[\&, \#, \$, @, \%, *]$.

Which of the following statements is/are MUST be true?

A. $\&$ is the smallest element of the array.
B. $*$ is the largest element of the array.
C. $\& < \#$
D. $\% < @$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

In a Binary Search Tree (BST) containing $15$ nodes, what is the maximum depth of the median value in the tree?

Your Answer: | Correct Answer: 14 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following Python function:

```
    def z(n):
        if n <= 10:
            return
        elif n % 2 == 0:
5.          z(n // 2)
            z(n // 2)
        else:
            z(n + 1)
```

The function recursively processes the input $n$, with two recursive calls for even numbers and one recursive call for odd numbers.

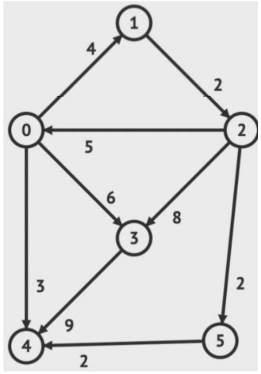
What is the runtime for the function $z(n)$?

A. $\theta\left((\log N)^2\right)$
B. $\theta(\log N)$
C. $\theta(N)$
D. $\theta(N \log N)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Suppose we run Dijkstra's algorithm on the graph below, starting from vertex $2$. In what order will the vertices be removed from the priority queue during the execution of the algorithm?




A. $0, 4, 1, 3, 5, 2$
B. $2, 0, 4, 1, 3, 5$
C. $2, 0, 3, 1, 5, 4$
D. $2, 5, 4, 0, 3, 1$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider the following Python method implemented for a Queue class, where the queue stores integer values:

```
    def mystery(q, x):
        if q.is_empty():
            q.enqueue(x)
            return
5.      for _ in range(q.size()):
            y = q.peek() # return the front element of the queue without removing it
            if y == x:
                return
            q.dequeue()
10.         q.enqueue(y)
```

What does this method do to the queue q?

A. Inserts $x$ into the queue as the first element if the queue is empty, otherwise leaves the queue unchanged.
B. Rotates the queue's elements until $x$ is at the front of the queue or leaves the queue unchanged if $x$ is not found.
C. Inserts $x$ into the queue only if it is not already present, ensuring no duplicates of $x$.
D. Completely reverses the order of the queue's elements.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

If $A$ is a $10 \times 3$ matrix with an SVD $U\Sigma V^T$, which of the following correctly describe the projection matrices onto $C(A)$ and $C\left(A^T\right)$?

A. Projection onto $C(A)$ is $UU^T$, and projection onto $C\left(A^T\right)$ is $VV^T$.
B. Projection onto $C(A)$ is $UU^T$, and projection onto $C\left(A^T\right)$ is $I$.
C. Projection onto $C(A)$ is $VV^T$, and projection onto $C\left(A^T\right)$ is $UU^T$.
D. Projection onto $C(A)$ is $I$, and projection onto $C\left(A^T\right)$ is $VV^T$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Let $G$ be a graph, and suppose $s$, $u$, and $v$ are three nodes in the graph. A depth-first search (DFS) is run on $G$ starting from $s$, with neighbors processed in ascending order. It is found that:

$$\text{start}[u] < \text{start}[v] < \text{finish}[v] < \text{finish}[u]$$

Now, suppose another DFS is run starting from node $s$, but with a different order of processing neighbors. Let new_start and new_finish represent the start and finish times from this second DFS.

Which of the following statement(s) MAY BE correct? Select all that apply.

A. new_start $[v]$ < new_start $[u]$ < new_finish $[u]$ < new_finish $[v]$
B. new_start $[u]$ < new_start $[v]$ < new_finish $[v]$ < new_finish $[u]$

C. new_start $[v]$ < new_finish $[v]$ < new_start $[u]$ < new_finish $[u]$
D. new_start $[u]$ < new_finish $[u]$ < new_start $[v]$ < new_finish $[v]$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Determine which of the following statements about graph properties is/are TRUE?

A. An undirected graph G with $|\mathrm{V}| = |\mathrm{E}|$ is necessarily a tree.
B. Depth First Search (DFS) on an undirected graph produces no cross edges in the resulting DFS tree.
C. If a DFS tree contains no back edges, the graph is acyclic.
D. Any tree with n vertices always has exactly $\mathrm{n} - 1$ edges.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following statements:

1. If $f(n) = O(g(n))$ and $\log g(n) \geq 1$ for all $n$, then $\log f(n) = O(\log g(n))$.
2. If $f(n) = O(g(n))$, then $2^{f(n)} = O\left(2^{g(n)}\right)$.

Which of the following is true?

A. Both statements are TRUE.
B. Statement 1 is TRUE, and Statement 2 is FALSE.
C. Statement 1 is FALSE, and Statement 2 is TRUE.
D. Both statements are FALSE.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

I have a bag with 3 coins in it. One of them is a fair coin, but the others are biased trick coins. When flipped, the three coins come up heads with probabilities $0.5, 0.6$, and 0.1, respectively. Suppose that I pick one of these three coins uniformly at random and flip it two times.

What is $\mathrm{P}(\mathrm{HT})$ ? (That is, the coin comes up heads on the first flip and tails on the second flip.)

A. $55/300$
B. $53/300$
C. $57/300$
D. $58/300$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

There are 8 cards in a hat:

$$\{1\heartsuit, 1\spadesuit, 1\diamondsuit, 1\clubsuit, 2\heartsuit, 2\spadesuit, 2\diamondsuit, 2\clubsuit\}$$

You draw one card at random.

- If its rank is 1, you draw one more card.
- If its rank is 2, you draw two more cards.

Let $X$ be the sum of the ranks on the 2 or 3 cards drawn. What is the expected value $E(X)$?

A. $24/7$
B. $26/7$
C. $28/7$
D. $30/7$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $Y_1$ and $Y_2$ be jointly continuous random variables with the joint density function:

$$f(y_1, y_2) = \begin{cases} cy_2 & \text{if } 0 \leq y_1 \leq 1 \text{ and } y_1^2 \leq y_2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

What is the value of $c$?

A. $2/5$
B. $5/2$
C. $3/2$
D. $1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose 10 balls are thrown into 10 bins, where each ball is equally likely to land in any one of the 10 bins. The bins are numbered 1 to 10. Let:

- $X_1$ : Number of balls that land in bin 1.
- $X_2$ : Number of balls that land in bin 2.
- Similarly, $X_3, X_4, \ldots, X_{10}$.

Define $Z$ as the maximum number of balls in any single bin:
$Z = \max(X_1, X_2, \ldots, X_{10})$.

What is the probability that $Z = 1$?

A. $\frac{9!}{10^{10}}$

B. $\frac{10!}{10^{10}}$

C. $\frac{10}{10^{10}}$

D. $\frac{1}{10}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

An insurance company determines that $N$, the number of claims received in a week, is a random variable with $P[N = n] = \frac{1}{2^{n+1}}$ where $n \geq 0$. The company also determines that the number of claims received in a given week is independent of the number of claims received in any other week.

Calculate the probability that exactly seven claims will be received during a given two-week period.

A. $1/256$
B. $1/128$
C. $7/512$
D. $1/64$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Assume a data set of 4 points $(1, 1), (2, 4), (4, 2), (5, 5)$ in $\mathbb{R}^2$. What is the first principal direction?

A. $\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

B. $\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

C. $\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ -1 \end{bmatrix}$

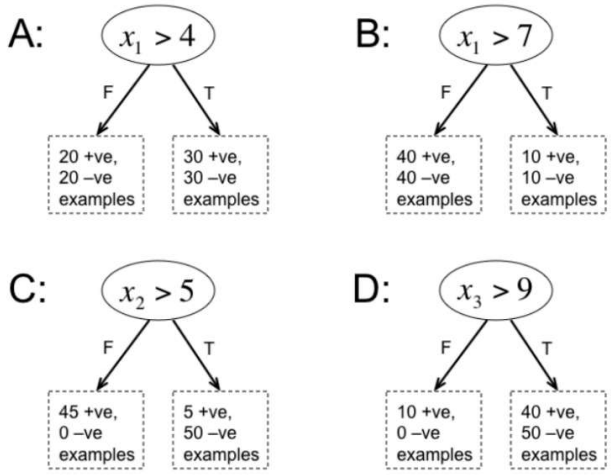
D. $\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Suppose you have a binary classification problem with 3-dimensional feature vectors $\mathbf{x} \in \mathbb{R}^3$. You are given 50 positive and 50 negative training examples and want to build a decision tree classifier. Consider 4 possible splits at the root node.

Which of the following splits gives the highest information gain?




Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a dataset with seven points $\{x_1, \ldots, x_7\}$. Given below are the distances between all pairs of points.

| | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ | $x_7$ |
|---|---|---|---|---|---|---|---|
| $x_1$ | 0 | 5 | 3 | 1 | 6 | 2 | 3 |
| $x_2$ | 5 | 0 | 4 | 6 | 1 | 7 | 8 |
| $x_3$ | 3 | 4 | 0 | 4 | 3 | 5 | 6 |
| $x_4$ | 1 | 6 | 4 | 0 | 7 | 1 | 2 |
| $x_5$ | 6 | 1 | 3 | 7 | 0 | 8 | 9 |
| $x_6$ | 2 | 7 | 5 | 1 | 8 | 0 | 1 |
| $x_7$ | 3 | 8 | 6 | 2 | 9 | 1 | 0 |

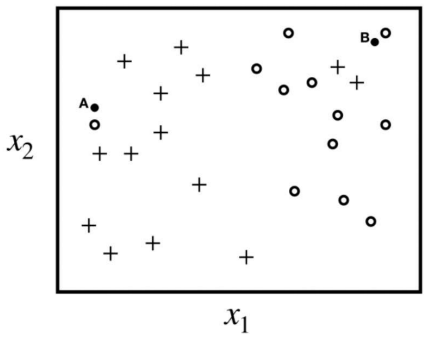
Assume that $k = 2$, and the cluster centers are initialized to $x_3$ and $x_6$. Which of the following shows the two clusters formed at the end of the first iteration of k -k-means? Circle the correct option.

A. $\{x_1, x_2, x_3, x_4\}, \{x_5, x_6, x_7\}$
B. $\{x_2, x_3, x_5\}, \{x_1, x_4, x_6, x_7\}$
C. $\{x_1, x_2, x_3, x_5\}, \{x_4, x_6, x_7\}$
D. $\{x_2, x_3, x_4, x_7\}, \{x_1, x_5, x_6\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a binary classification dataset with two-dimensional input vectors ( $\mathrm{x1}, \mathrm{x2}$ ) and labels + and ○. Two new points, $\mathbf{A}$ and $\mathbf{B}$, are introduced for classification. Assume the K-Nearest Neighbors (K-NN) algorithm uses Euclidean distance to make predictions.



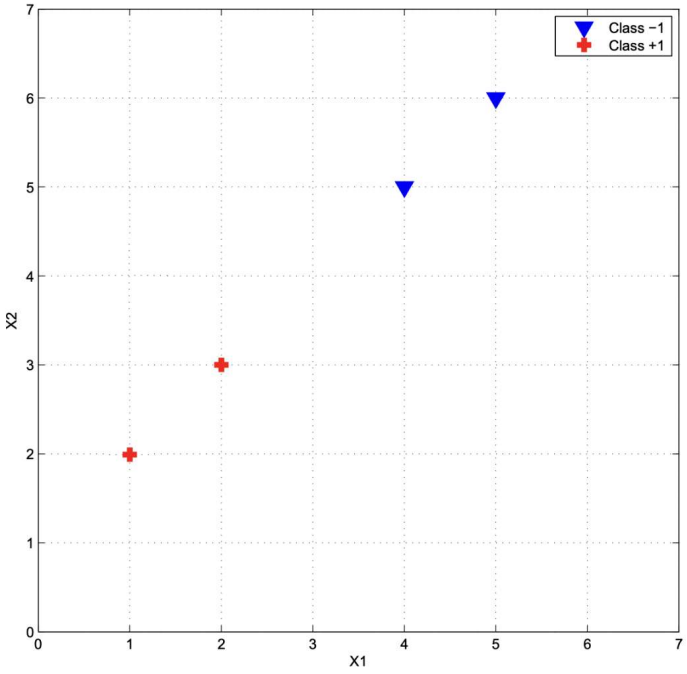

What are the predicted labels for points $\mathbf{A}$ and $\mathbf{B}$ under the 3-NN classifier?

A. 3-NN predicts + for A and + for B.
B. 3-NN predicts + for A and ○ for B
C. 3-NN predicts ○ for A and + for B.
D. 3-NN predicts ○ for A and ○ for B.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Suppose we train a hard margin SVM on the given dataset (as shown in the plot).




What will be the weight vector $\mathrm{w}_1, \mathrm{w}_2$ and bias b values returned by the SVM?

A. $w1 = w2 = -1/2$ and $b = 7/2$
B. $w1 = w2 = -1$ and $b = 7$
C. $w1 = w2 = 1$ and $b = 7$
D. $\mathrm{w1} = \mathrm{w2} = 1/2$ and $b = -7/2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Consider the function $f(x, y) = 2x^3 + xy^2 + 5x^2 + y^2$. Which statement is true?

A. The function $f$ has both a local minimum and a local maximum point.
B. The function $f$ has a local minimum but no local maximum point.
C. The function $f$ has a local maximum but no local minimum point.
D. All stationary points of $f$ are saddle points.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Suppose that two relations $R(A, B)$ and $S(A, B)$ have exactly the same schema. Consider the following equalities in relational algebra?

I. $R \cap S = R - (R - S)$
II. $R \cap S = S - (S - R)$
III. $R \cap S = R$ NATURAL-JOIN $S$
IV. $\mathrm{R} \cap \mathrm{S} = \mathrm{R} \times \mathrm{S}$

Which of the equalities hold in relational algebra?

A. I only
B. I and II only
C. I, II and III
D. I, II, III and IV

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider a relation $\mathrm{R}(\mathrm{A}, \mathrm{B})$ that contains $100$ tuples, and a relation $\mathrm{S}(\mathrm{B}, \mathrm{C})$ that contains $80$ tuples; Make no assumptions about keys $\&$ assume that the domain of all attributes is integers. Consider the following relational algebra expression:

$$\pi_{\mathrm{B}}\mathrm{R} - (\pi_{\mathrm{B}}\mathrm{R} - \pi_{\mathrm{B}}\mathrm{S})$$

Let $m$ be the minimum and $\mathrm{M}$ be the maximum number of tuples that could be in the result of the above expression, then $m + \mathrm{M}$?

Your Answer: | Correct Answer: 80 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Numerical Type | Award: 2 | Penalty: 0 | Databases

In the context of relational database model and relational algebra, Let $\mathrm{A}, \mathrm{B}$ be two relation schemes.

For any relation instances $a(\mathrm{A}), b(\mathrm{B})$; We say that a relation instance $\mathrm{Y}$ is the Yield of $a, b$ if $\mathrm{Y}$ is the largest relation instance such that $\mathrm{Y} \times b \subseteq a$ (Note that " $\times$ " is the cross product).

Consider the following relation instances $r, s$ :

$r$

| A | B | C | D | E |
|---|---|---|---|---|
| $\alpha$ | a | $\alpha$ | a | 1 |
| $\alpha$ | a | $\gamma$ | a | 1 |
| $\alpha$ | a | $\gamma$ | b | 1 |
| $\beta$ | a | $\gamma$ | a | 1 |
| $\beta$ | a | $\gamma$ | b | 3 |
| $\gamma$ | a | $\gamma$ | a | 1 |
| $\gamma$ | a | $\gamma$ | b | 1 |
| $\gamma$ | a | $\beta$ | b | 1 |

$s$

| D | E |
|---|---|
| a | 1 |
| b | 1 |

The number of tuples in the Yield of $r$, $s$ is _____

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Given the Bayes net shown in the Figure below; A, B, C, D, and E are all Boolean variables. $\mathrm{P}(\mathrm{A} =" + ")$ is simply denoted as $\mathrm{P}(\mathrm{A})$.

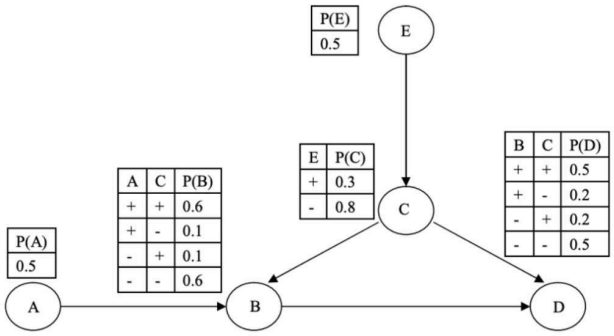



Figure 2: Bayes Net

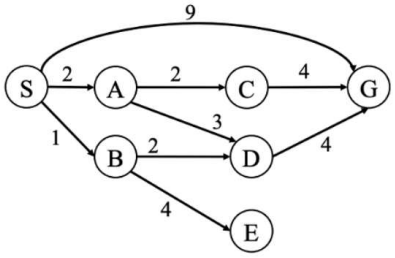
Note: In the following, we'll use the notations: $A \perp B$ means A is independent of B; $A \perp B \mid C$ means A is conditionally independent of B given C.

Which of the following independence assumptions are correct?

A. $B \perp E \mid C$
B. $A \perp D$
C. $A \perp D \mid B$
D. $A \perp D \mid B, C$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence




Heuristic

| S | A | B | C | D | E | G |
|---|---|---|---|---|---|---|
| 6 | 0 | 6 | 4 | 1 | 10 | 0 |

What path will $A^*$ graph search return?

A. $S - A - C - G$
B. $S - B - D - G$
C. $S - A - D - G$
D. $S - G$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Consider the following Python code:

```
    def snake(M):
        r = len(M)
        c = len(M[0])
        ans = []

        for i in range(r):
            if i % 2 == 0:
                for j in M[i]:
                    ans.append(j)
        else:
                l = M[i]
                for j in range(-1, -c - 1, -1):
                    ans.append(l[j])

    return ans
```

What is the output of the program when executed with the input:

$M = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]$?

A. $[1, 2, 3, 4, 5, 6, 7, 8, 9]$
B. $[1, 2, 3, 6, 5, 4, 7, 8, 9]$
C. $[1, 2, 3, 9, 8, 7, 6, 5, 4]$
D. $[3, 2, 1, 6, 5, 4, 9, 8, 7]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following sets represent vector subspaces of $\mathbb{R}^3$?

A. $\mathcal{S} = \{(x, y, z)^T \mid x + y + z + 1 = 0\}$
B. $\mathcal{S} = \{(x, y, z)^T \mid x \geq y \geq z \geq 0\}$
C. $\mathcal{S} = \{(x, y, z)^T \mid x - y + z = 0\}$
D. $\mathcal{S} = \{(x, y, z)^T \mid x - y = 0 \text{ and } z = 0\}$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let $W$ be a subspace of $\mathbb{R}_3$ spanned by $(1, 2, 3), (2, k, 3), (4, 5, k + 8)$. Determine the values of $k$ so that $W^\perp$ has dimension zero.

A. $k \neq 7$
B. $k \neq 7, k \neq -1$
C. $k \neq 7$ and $k \neq 1$
D. $k = 7, k = 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Suppose that $A$ is a $4 \times 4$ real matrix with four orthogonal but not orthonormal columns $a_i$ with lengths $\|a_1\| = 2, \|a_2\| = 1, \|a_3\| = 3, \|a_4\| = 2$ (i.e., $a_i^T a_j = 0$ for $i \neq j$ ). Which of the following represents the correct diagonal matrix $\Sigma$ in the singular value decomposition $A = U\Sigma V^T$?

A. $\Sigma = \text{diag}(3, 2, 2, 1)$
B. $\Sigma = \text{diag}(3, 3, 2, 1)$
C. $\Sigma = \text{diag}(3, 2, 1, 1)$
D. $\Sigma = \text{diag}(3, 3, 2, 2)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let $A$ and $B$ be two real $m \times m$ matrices, where $B$ is invertible. Select all statements that are necessarily true.

A. $\text{Col}(A) = \text{Col}(AB)$
B. $\text{Col}(A) = \text{Col}(BA)$
C. $\text{Null}(A) = \text{Null}(AB)$
D. $\text{Null}(A) = \text{Null}(BA)$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

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